JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Trigonometric Equations

JEE Main Mathematics · Previous Year Questions with Answer Key

62	61	2002 – 2026	40	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 18 questions

Q1. JEE Main 2026 (Online) 6th April Morning Shift

Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$.

Then $\sum\limits_{\theta\in S} \theta$ is equal to :

(A) $-\frac{2\pi}{3}$ (B) $-\frac{4\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{4\pi}{3}$

Q2. JEE Main 2026 (Online) 5th April Morning Shift

The sum of all the integral values of p such that the equation $3\sin^2 x + 12\cos x - 3 = p, x \in \mathbb{R}$, has at least one solution, is:

(A) -54 (B) -60 (C) -75 (D) -84

Q3. JEE Main 2026 (Online) 2nd April Morning Shift

Let $S = \{x \in [-\pi, \pi] : \sin x(\sin x + \cos x) = a,\ a \in \mathbb{Z}\}$. Then $n(S)$ is equal to:

(A) 3 (B) 6 (C) 7 (D) 9

Q4. JEE Main 2026 (Online) 23rd January Morning Shift

Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0, \theta \in [-3\pi, 2\pi]$ is :

(A) 5 (B) 4 (C) 3 (D) 0

Q5. JEE Main 2025 (Online) 7th April Evening Shift

The number of solutions of the equation

$\cos 2\theta \cos\frac{\theta}{2} + \cos\frac{5\theta}{2} = 2\cos^3\frac{5\theta}{2}$ in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is :

(A) 5 (B) 7 (C) 6 (D) 9

Q6. JEE Main 2025 (Online) 3rd April Evening Shift

The number of solutions of the equation

$(4-\sqrt{3})\sin x - 2\sqrt{3}\cos^2 x = -\frac{4}{1+\sqrt{3}}, x \in \left[-2\pi, \frac{5\pi}{2}\right]$ is

(A) 4 (B) 3 (C) 6 (D) 5

Q7. JEE Main 2025 (Online) 3rd April Morning Shift

The number of solutions of the equation $2x + 3\tan x = \pi, x \in [-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$ is:

(A) 4 (B) 5 (C) 3 (D) 6

Q8. JEE Main 2025 (Online) 2nd April Evening Shift

If $\theta \epsilon \left[-\frac{7\pi}{6}, \frac{4\pi}{3}\right]$, then the number of solutions of $\sqrt{3}\,\text{cosec}^2\,\theta - 2(\sqrt{3}-1)\,\text{cosec}\,\theta - 4 = 0$, is equal to :

(A) 7 (B) 10 (C) 6 (D) 8

Q9. JEE Main 2025 (Online) 2nd April Morning Shift

If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2}\cos^2\theta + (2-\sqrt{6})\cos\theta - \sqrt{3} = 0$, is equal to:

(A) 8 (B) 6 (C) 10 (D) 12

Q10. JEE Main 2025 (Online) 22nd January Evening Shift

The sum of all values of $\theta \in [0, 2\pi]$ satisfying $2\sin^2\theta = \cos 2\theta$ and $2\cos^2\theta = 3\sin\theta$ is

(A) π (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{2}$ (D) 4π

Q11. JEE Main 2024 (Online) 9th April Morning Shift

Let $|\cos\theta\cos(60-\theta)\cos(60+\theta)| \leq \frac{1}{8}, \theta \in [0, 2\pi]$. Then, the sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value, is :

(A) 6π (B) 9π (C) 18π (D) 15π

Q12. JEE Main 2024 (Online) 5th April Morning Shift

Suppose $\theta \in \left[0, \frac{\pi}{4}\right]$ is a solution of $4\cos\theta - 3\sin\theta = 1$. Then $\cos\theta$ is equal to :

(A) $\frac{6-\sqrt{6}}{(3\sqrt{6}-2)}$ (B) $\frac{4}{(3\sqrt{6}+2)}$

(C) $\frac{6+\sqrt{6}}{(3\sqrt{6}+2)}$ (D) $\frac{4}{(3\sqrt{6}-2)}$

Q13. JEE Main 2024 (Online) 1st February Evening Shift

The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ is :

(A) 0 (B) 3 (C) 1 (D) 2

Q14. JEE Main 2024 (Online) 30th January Morning Shift

If $2\sin^3 x + \sin 2x\cos x + 4\sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right], n \in \mathrm{N}$, then the roots of the equation $x^2 + nx + (n-3) = 0$ belong to :

(A) $(0, \infty)$ (B) Z

(C) $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$ (D) $(-\infty, 0)$

Q15. JEE Main 2024 (Online) 29th January Evening Shift

The sum of the solutions $x \in \mathbb{R}$ of the equation $\frac{3\cos 2x+\cos^3 2x}{\cos^6 x-\sin^6 x} = x^3 - x^2 + 6$ is

(A) 3 (B) 1 (C) 0 (D) -1

Q16. JEE Main 2024 (Online) 29th January Morning Shift

If $\alpha, -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ is the solution of $4\cos\theta + 5\sin\theta = 1$, then the value of $\tan\alpha$ is

(A) $\frac{10-\sqrt{10}}{12}$ (B) $\frac{\sqrt{10}-10}{6}$ (C) $\frac{\sqrt{10}-10}{12}$ (D) $\frac{10-\sqrt{10}}{6}$

Q17. JEE Main 2024 (Online) 27th January Evening Shift

If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in \mathbf{N}$, then $\sum_{k=1}^{n} \frac{k}{2^k}$ is equal to:

(A) $\frac{1}{2^{14}}\left(2^{15}-15\right)$ (B) $1 - \frac{15}{2^{13}}$

(C) $\frac{1}{2^{15}}\left(2^{14}-14\right)$ (D) $\frac{1}{2^{13}}\left(2^{14}-15\right)$

Q18. JEE Main 2016 (Online) 10th April Morning Slot

Let P = $\{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$

and Q = $\{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. Then

(A) P $\subset$ Q and Q $-$ P $\neq \phi$ (B) Q $\not\subset$ P

(C) P $\not\subset$ Q (D) P = Q

Numerical 4 questions

Q19. JEE Main 2026 (Online) 5th April Evening Shift

If $S = \left\{\theta \in [-\pi, \pi] : \cos\theta\cos\frac{5\theta}{2} = \cos 7\theta\cos\frac{7\theta}{2}\right\}$, then $n(S)$ is equal to ____ .

Q20. JEE Main 2026 (Online) 24th January Evening Shift

The number of elements in the set $\{x \in [0, 180^\circ] : \tan(x+100^\circ) = \tan(x+50^\circ)\tan x\tan(x-50^\circ)\}$ is ____ .

Q21. JEE Main 2024 (Online) 5th April Evening Shift

The number of solutions of $\sin^2 x + \left(2 + 2x - x^2\right) \sin x - 3(x-1)^2 = 0$, where $-\pi \leq x \leq \pi$, is _____.

Q22. JEE Main 2024 (Online) 4th April Evening Shift

Let $S = \{\sin^2 2\theta : \left(\sin^4 \theta + \cos^4 \theta\right)x^2 + (\sin 2\theta)x + \left(\sin^6 \theta + \cos^6 \theta\right) = 0$ has real roots $\}$. If α and β be the smallest and largest elements of the set S, respectively, then $3\left((\alpha-2)^2 + (\beta-1)^2\right)$ equals _____.

Out of syllabus 40 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q23. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

The number of elements in the set

$S = \left\{\theta \in [0, 2\pi] : 3\cos^4 \theta - 5\cos^2 \theta - 2\sin^6 \theta + 2 = 0\right\}$ is :

(A) 9 (B) 8 (C) 12 (D) 10

Q24. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{\tan^2 x} = 10\right\}$ and $\beta = \sum\limits_{x \in S} \tan^2\left(\frac{x}{3}\right)$, then $\frac{1}{6}(\beta - 14)^2$ is equal to :

(A) 16 (B) 32 (C) 8 (D) 64

Q25. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

The number of elements in the set $S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2+x}{6}\right) = 4^x + 4^{-x}\right\}$ is :

(A) 1 (B) 3 (C) 0 (D) infinite

Q26. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Let $S = \left\{\theta \in \left(0, \frac{\pi}{2}\right) : \sum\limits_{m=1}^{9} \sec\left(\theta + (m-1)\frac{\pi}{6}\right) \sec\left(\theta + \frac{m\pi}{6}\right) = -\frac{8}{\sqrt{3}}\right\}$. Then

(A) $S = \left\{\frac{\pi}{12}\right\}$ (B) $S = \left\{\frac{2\pi}{3}\right\}$

(C) $\sum\limits_{\theta \in S} \theta = \frac{\pi}{2}$ (D) $\sum\limits_{\theta \in S} \theta = \frac{3\pi}{4}$

Q27. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

Let $S = \left\{\theta \in [0, 2\pi] : 8^{2\sin^2\theta} + 8^{2\cos^2\theta} = 16\right\}$. Then $n(S) + \sum\limits_{\theta \in S} \left(\sec\left(\frac{\pi}{4} + 2\theta\right) \operatorname{cosec}\left(\frac{\pi}{4} + 2\theta\right)\right)$ is equal to:

(A) 0 (B) -2 (C) -4 (D) 12

Q28. JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

The number of solutions of $|\cos x| = \sin x$, such that $-4\pi \le x \le 4\pi$ is :

(A) 4 (B) 6 (C) 8 (D) 12

Q29. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

The number of solutions of the equation

$\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is :

(A) 8 (B) 5 (C) 6 (D) 7

Q30. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

Let $S = \left\{\theta \in [-\pi, \pi] - \left\{\pm\frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$.

If $T = \sum\limits_{\theta \in S} \cos 2\theta$, then T + n(S) is equal to :

(A) $7 + \sqrt{3}$ (B) 9 (C) $8 + \sqrt{3}$ (D) 10

Q31. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

If n is the number of solutions of the equation
$2\cos x\left(4\sin\left(\frac{\pi}{4} + x\right)\sin\left(\frac{\pi}{4} - x\right) - 1\right) = 1, x \in [0, \pi]$ and S is the sum of all these solutions, then the ordered pair (n, S) is :

(A) (3, 13π / 9) (B) (2, 2π / 3) (C) (2, 8π / 9) (D) (3, 5π / 3)

Q32. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

The number of solutions of the equation $32^{\tan^2 x} + 32^{\sec^2 x} = 81$, $0 \le x \le \frac{\pi}{4}$ is :

(A) 3 (B) 1 (C) 0 (D) 2

Q33. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

The sum of solutions of the equation

$\frac{\cos x}{1+\sin x} = |\tan 2x|$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}, -\frac{\pi}{4}\right\}$ is :

(A) $-\frac{11\pi}{30}$ (B) $\frac{\pi}{10}$ (C) $-\frac{7\pi}{30}$ (D) $-\frac{\pi}{15}$

Q34. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

The sum of all values of x in $[0, 2\pi]$, for which $\sin x + \sin 2x + \sin 3x + \sin 4x = 0$, is equal to :

(A) 8π (B) 11π (C) 12π (D) 9π

Q35. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

The number of solutions of $\sin^7 x + \cos^7 x = 1$, $x \in [0, 4\pi]$ is equal to

(A) 11 (B) 7 (C) 5 (D) 9

Q36. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

The number of solutions of the equation $x + 2\tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is :

(A) 4 (B) 3 (C) 2 (D) 5

Q37. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

The number of roots of the equation, $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ in the interval $[0, \pi]$ is equal to :

(A) 2 (B) 3 (C) 4 (D) 8

Q38. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in :

(A) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

(B) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(C) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(D) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

Q39. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

Let S be the set of all $\alpha \in R$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to :

(A) [2, 6] (B) [3, 7] (C) [1, 4] (D) R

Q40. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

The number of solutions of the equation
$1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is :

(A) 5 (B) 3 (C) 7 (D) 4

Q41. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Let S = $\{\theta \in [-2\pi, 2\pi] : 2\cos^2\theta + 3\sin\theta = 0\}$. Then the sum of the elements of S is

(A) π (B) 2π (C) $\frac{13\pi}{6}$ (D) $\frac{5\pi}{3}$

Q42. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfying
$\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is -

(A) $\frac{5\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) $\frac{3\pi}{8}$

Q43. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

If $0 \leq x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is :

(A) 3 (B) 1 (C) 4 (D) 2

Q44. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

If sum of all the solutions of the equation

$$8\cos x.\left(\cos\left(\frac{\pi}{6}+x\right).\cos\left(\frac{\pi}{6}-x\right)-\frac{1}{2}\right)=1$$

in $[0, \pi]$ is $k\pi$, then k is equal to

(A) $\frac{20}{9}$ (B) $\frac{2}{3}$ (C) $\frac{13}{9}$ (D) $\frac{8}{9}$

Q45. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

The number of solutions of $\sin 3x = \cos 2x$, in the interval $\left(\frac{\pi}{2}, \pi\right)$ is :

(A) 1 (B) 2 (C) 3 (D) 4

Q46. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The number of $x \in [0, 2\pi]$ for which

$$\left|\sqrt{2\sin^4 x + 18\cos^2 x} - \sqrt{2\cos^4 x + 18\sin^2 x}\right| = 1$$ is :

(A) 2 (B) 4 (C) 6 (D) 8

Q47. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

If $0 \leq x < 2\pi$, then the number of real values of x, which satisfy the equation
$\cos x + \cos 2x + \cos 3x + \cos 4x = 0$ is:

(A) 7 (B) 9 (C) 3 (D) 5

Q48. AIEEE 2006 · MCQ (Single Correct Answer)

The number of values of x in the interval $[0, 3\pi]$ satisfying the equation $2\sin^2 x + 5\sin x - 3 = 0$ is

(A) 4 (B) 6 (C) 1 (D) 2

Q49. AIEEE 2002 · MCQ (Single Correct Answer)

The number of solution of $\tan\, x + \sec\, x = 2\cos\, x$ in $[0,\, 2\,\pi]$ is

(A) 2 (B) 3 (C) 0 (D) 1

Q50. JEE Main 2023 (Online) 25th January Evening Shift · Numerical

If m and n respectively are the numbers of positive and negative values of θ in the interval $[-\pi, \pi]$ that satisfy the equation $\cos 2\theta \cos\frac{\theta}{2} = \cos 3\theta \cos\frac{9\theta}{2}$, then mn is equal to _____.

Q51. JEE Main 2023 (Online) 24th January Evening Shift · Numerical

Let $S = \{\theta \in [0, 2\pi) : \tan(\pi\cos\theta) + \tan(\pi\sin\theta) = 0\}$. Then $\sum_{\theta\in S} \sin^2\left(\theta + \frac{\pi}{4}\right)$ is equal to _____.

Q52. JEE Main 2022 (Online) 29th July Morning Shift · Numerical

Let $S = \left\{\theta \in (0, 2\pi) : 7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2\right\}$. Then, the sum of roots of all the equations $x^2 - 2\left(\tan^2\theta + \cot^2\theta\right)x + 6\sin^2\theta = 0, \theta \in S$, is _____.

Q53. JEE Main 2022 (Online) 28th July Evening Shift · Numerical

Let $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$. Then the number of elements in the set $A = \{\theta \in S : \tan\theta(1 + \sqrt{5}\tan(2\theta)) = \sqrt{5} - \tan(2\theta)\}$ is _____.

Q54. JEE Main 2022 (Online) 26th July Evening Shift · Numerical

If the sum of solutions of the system of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta + 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is $k\pi$, then k is equal to _____.

Q55. JEE Main 2022 (Online) 30th June Morning Shift · Numerical

Let $S_1 = \{x \in [0, 12\pi] : \sin^5 x + \cos^5 x = 1\}$

and $S_2 = \{x \in [0, 8\pi] : \sin^7 x + \cos^7 x = 1\}$

Then $n(S_1) - n(S_2)$ is equal to _____.

Q56. JEE Main 2022 (Online) 29th June Evening Shift · **Numerical**

The number of solutions of the equation $\sin x = \cos^2 x$ in the interval (0, 10) is _____.

Q57. JEE Main 2022 (Online) 29th June Morning Shift · **Numerical**

The number of elements in the set $S = \{\theta \in [-4\pi, 4\pi] : 3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0\}$ is _____.

Q58. JEE Main 2022 (Online) 29th June Morning Shift · **Numerical**

The number of solutions of the equation

$2\theta - \cos^2\theta + \sqrt{2} = 0$ in R is equal to _____.

Q59. JEE Main 2022 (Online) 25th June Morning Shift · **Numerical**

The number of values of x in the interval $\left(\frac{\pi}{4}, \frac{7\pi}{4}\right)$ for which

$14\cos ec^2 x - 2\sin^2 x = 21 - 4\cos^2 x$ holds, is _____.

Q60. JEE Main 2021 (Online) 27th August Evening Shift · **Numerical**

Let S be the sum of all solutions (in radians) of the equation $\sin^4\theta + \cos^4\theta - \sin\theta\cos\theta = 0$ in [0, 4π]. Then $\frac{8S}{\pi}$ is equal to _____.

Q61. JEE Main 2021 (Online) 18th March Morning Shift · **Numerical**

The number of solutions of the equation

$|\cot x| = \cot x + \frac{1}{\sin x}$ in the interval [0, 2π] is

Q62. JEE Main 2021 (Online) 26th February Morning Shift · **Numerical**

If $\sqrt{3}(\cos^2 x) = (\sqrt{3} - 1)\cos x + 1$, the number of solutions of the given equation when $x \in \left[0, \frac{\pi}{2}\right]$ is _____.

Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **D**	4. **A**	5. **B**
6. **D**	7. **B**	8. **C**	9. **A**	10. **A**
11. **A**	12. **D**	13. **A**	14. **D**	15. **D**
16. **C**	17. **D**	18. **D**		

Numerical

19. **19**	20. **4**	21. **2**	22. **4**	

Out of syllabus

23. **A**	24. **B**	25. **A**	26. **C**	27. **C**
28. **C**	29. **D**	30. **B**	31. **A**	32. **B**
33. **A**	34. **D**	35. **C**	36. **B**	37. **C**
38. **D**	39. **A**	40. **A**	41. **B**	42. **B**
43. **D**	44. **C**	45. **A**	46. **D**	47. **A**
48. **A**	49. **B**	50. **25**	51. **2**	52. **16**
53. **5**	54. **3**	55. **4**	56. **4**	57. **32**
58. **1**	59. **4**	60. **56**	61. **1**	62. **1**